

PinIT Career OS — 36+ Courses Master Visuals & Duration Architecture

Document Identifier

PINIT_36_INDIVIDUAL_COURSES_VISUALS_PLAN (v3.0 Zero-Hallucination & Duration Audit)

Total Courses & Volume

37 Master Courses • 1,110 Lesson Days • 6,660 Interactive Lesson Parts

Pedagogical Standard

18–25 Minutes Lesson Duration • Deterministic Gate v2 Visual Bindings

Actual Measured Reality

13 Courses Already Built (30/30) • 24 Courses Standalone (Expansion + Visuals Needed)

1. Honest Reality Audit: Already Built vs. To Be Built

An audit of the codebase confirms that some courses already have 30-day visual suites built from the Python and Web certification tracks. The 37 courses are structured into 4 reality categories:

Category	Description	Count	Visuals Status	Lesson Duration
Category A: Ready to Ship	Already built in cert tracks + LongLessons format	12 Courses	✓ 30/30 Built	🎯 21.2–24.4m (PERFECT)
Category B: Minor Trim Needed	Visuals 30/30 built (<code>sql-mastery</code>), duration slightly high	1 Course	✓ 30/30 Built	⚠️ 26.1m (Trim to 23m)
Category C: Visuals Needed	Duration perfect (<code>design</code>), but visuals need authoring	1 Course	⌚ 0/30 Needed	🎯 22.5m (PERFECT)
Category D: Standalone Courses	Standalone courses in short format (~6m); needs expansion & visuals	23 Courses	⌚ 0/30 Needed	⚠️ 6.0m (Expand to 18-25m)

2. The 18–25 Minute Pedagogical Duration Standard

In PinIT Career OS, lesson duration is calculated using the platform's official formula in `src/lib/data/longLessons.ts` :

Total Duration = **Spoken Time** ($\text{words} \div 120 \text{ wpm}$) + **Hands-on Activity Time** ($6 \text{ parts} \times 2.0 \text{ min per try\& check}$).
Target: 1,200–1,500 words of spoken lecture (10–12.5m) + interactive coding changes and diagnostic checks (11–12m) = 21.0 to 24.5 minutes.

3. Master Course-by-Course Duration & Visuals Audit (All 37 Courses)

#	Course Identifier	Title	Spoken	Activity	Total	Visuals
01	<code>course-java-logic</code>	Java Fundamentals & Core Logic	1m	5.5m	6.5m	0/30
02	<code>course-react-web</code>	Full-Stack React Web Development	9.5m	11.9m	21.4m	30/30
03	<code>course-node-web</code>	Node.js & TypeScript Backend Engineering	11.6m	12m	23.6m	30/30
04	<code>course-cloud-native</code>	Cloud Native Architectures (AWS)	12.1m	12m	24.1m	30/30
05	<code>course-devops-cicd</code>	DevOps & CI/CD Pipeline Automation	10.6m	12m	22.6m	30/30
06	<code>course-design-systems</code>	UI/UX Design Systems & Visual Frontend	10.5m	12m	22.5m	0/30
07	<code>course-dsa-optim</code>	Data Structures & Algorithmic Optimizations	11.3m	12m	23.3m	30/30
08	<code>course-mobile-dev</code>	Mobile Application Development	1m	5m	6m	0/30

#	Course Identifier	Title	Spoken	Activity	Total	Visuals
09	course-cybersecurity	Cybersecurity Principles & Secure Systems	12.4m	12m	24.4m	30/30
10	course-database-eng	Database Engineering & Query Performance	14.1m	12m	26.1m	30/30
11	course-distributed-sys	High-Scale Distributed System Design	10.8m	12m	22.8m	30/30
12	course-ai-eng	AI Engineering & LLM Integration	10.6m	12m	22.6m	30/30
13	course-fullstack-js	Full-Stack JavaScript Engineering	1m	5m	6m	0/30
14	course-iot-embedded	IoT, Firmware & Embedded Systems	1m	5m	6m	0/30
15	course-3d-graphics	3D Interactive Graphics & Avatar Animation	1m	5m	6m	0/30
16	course-blockchain-web3	Blockchain, Web3 & Smart Contracts	1m	5m	6m	0/30
17	course-iot-network	IoT Wireless Networks & Protocols	1m	5m	6m	0/30
18	course-iot-edge-ai	Edge AI, DSP & TinyML Systems	1m	5m	6m	0/30
19	course-iot-security	Industrial IoT Security & Device Lifecycle	1m	5m	6m	0/30
20	course-python-backend	Python Programming & Backend Systems	10.6m	12m	22.6m	30/30
21	course-quant-systems	Quantitative Engineering & Low-Latency Trading Systems	9.4m	12m	21.4m	30/30
22	course-digital-accounting	Digital Accounting & Taxation (B.Com / BBA)	1m	5m	6m	0/30
23	course-finance-investment	Business Finance & Investment Management (B.Com / BBA)	1m	5m	6m	0/30
24	course-business-analytics	Business Analytics & Decision Intelligence (B.Com / BBA / MBA)	1m	5m	6m	0/30
25	course-marketing-branding	Marketing & Brand Management (B.Com / BBA / MBA)	1m	5m	6m	0/30
26	course-digital-marketing	Digital Marketing & Growth Strategy (B.Com / BBA / MBA)	1m	5m	6m	0/30
27	course-ecommerce-digital-biz	E-Commerce & Digital Business (B.Com / BBA / MBA)	1m	5m	6m	0/30
28	course-entrepreneurship-biz-mgmt	Entrepreneurship & Business Management (B.Com / BBA / MBA)	1m	5m	6m	0/30
29	course-sales-crm-success	Sales, Customer Success & CRM (B.Com / BBA / MBA)	1m	5m	6m	0/30
30	course-operations-supplychain-compliance	Operations, Supply Chain & Business Compliance (B.Com / BBA / MBA)	1m	5m	6m	0/30
31	course-ai-digital-transformation	AI & Digital Transformation for Business (B.Com / BBA / MBA)	1m	5m	6m	0/30
32	course-computer-fundamentals	Computer Literacy, Digital Productivity & OS Fundamentals	1m	5m	6m	0/30

#	Course Identifier	Title	Spoken	Activity	Total	Visuals
33	course-ai-prompt-literacy	Everyday AI Literacy & Prompt Engineering	9.2m	12m	21.2m	30/30
34	course-excel-data-viz	Excel & Data Analysis Fundamentals	1m	5m	6m	0/30
35	course-git-version-control	Git, GitHub & Version Control Basics	1m	5m	6m	0/30
36	course-softskills-communication	Professional Tech Communication & Interview Mastery	1m	5m	6m	0/30
37	course-nlp	Natural Language Processing & Computational Linguistics	9.5m	12m	21.5m	30/30

4. Master Visual Templates Catalog (17 Interactive Diagram Engines)

Template Key	Engine Name	Domain	Interactive Mechanics & Pedagogical Function
flow	Multi-Stage Flow	Universal	Linear and branching pipelines with stage status highlights
boxes	Memory & State Boxes	Systems / Web	Slots holding named variables, props, register values, or configs
table	Relational & Ledger Table	Data / Fin / API	Columns and rows with row-by-row or cell-by-cell step illumination
letters	Character Array	CompSci / NLP	Character cells with moving pointers for strings, tokens, and regex
compare	Comparative Panels	Universal	Side-by-side comparison with pass/fail indicators and pros/cons
cells	1D Memory Array Buffer	DSA / Hardware	Indexed array blocks with up to 3 named moving pointers
stack-queue	LIFO Stack / FIFO Queue	Systems / Trading	Vertical stack frames or horizontal queue with push/pop animations
tree-graph	Hierarchical Tree / Graph	DSA / Web / 3D	Up to 6 nodes with directional edges and traversal highlighting
bars	Quantitative Metrics Bars	Analytics / SRE	Labeled numeric bars comparing latency, ratios, or benchmarks
sequence	Chronological Sequence	Network / Auth	Lifeline columns with directional request/response message arrows
states	Finite State Machine	Protocols / Async	State bubbles with transitions and active state illumination
workflow	Multi-Tier Topology	Cloud / DevOps	System architecture topology connecting client, proxy, pods, DB
component-tree	UI Component Hierarchy	Frontend / Mobile	Component tree displaying props cascading down and events bubbling up
wireframe	Box-Model Wireframe	Frontend / Design	Nested UI boxes visualizing layout engines, margin, border, padding
register-bits	Bitwise Register Simulator	Embedded / IoT	8/16/32-bit register cells visualizing bitmasks, set/clear operations
ledger-sheet	Double-Entry T-Account	Commerce / Acct	T-Account debit/credit columns maintaining balancing invariants
funnel	Conversion Funnel	Marketing / Sales	Multi-stage conversion stages showing drop-off percentages and cohorts

5. Detailed Blueprint & Action Plan (All 37 Courses, One by One)

Below is the exhaustive architectural specification for every single course, specifying existing assets, measured times, allowed templates, and curriculum blocks:

Course 01: Java Fundamentals & Core Logic

Visuals: Needed (0/30) | Duration: 6.5 min (Expansion to 18-25m Needed) | Allowed: boxes, flow, table, cells, stack-queue, compare

course-java-logic

Architectural Theme: JVM Architecture, Primitive Typing, OOP Invariants & Enterprise Modularity

Pedagogical Whiteboard Model: Illustrate JVM Stack frame pushes/pops, heap reference allocations, and variable mutations during loops. Visuals clarify value vs reference semantics.

Block	Days	Curriculum Topic & Real Lessons	Visual Engine Architecture
Block 1	Days 1–5	JVM Bytecode execution, Primitive boxing & System.out printing flow	Template boxes : Animated step-by-step state progression
Block 2	Days 6–10	Conditional branching, Relational operators & Boolean logic tables	Template flow : Animated step-by-step state progression
Block 3	Days 11–15	While/Do-While loops, Iterative counters & Array indexing cells	Template table : Animated step-by-step state progression
Block 4	Days 16–20	For loops, Nested iteration patterns & Matrix coordinate grids	Template cells : Animated step-by-step state progression
Block 5	Days 21–25	Method call frames, Parameter pass-by-value & Return value routing	Template stack-queue : Animated step-by-step state progression
Block 6	Days 26–30	OOP Class blueprints, Encapsulation & Milestone Financial Engine	Template compare : Animated step-by-step state progression

Capstone Milestone: Milestone Financial Utility Engine: A multi-module Java application modeling bank accounts, compound interest calculators, and transactional debit/credit validation.

Course 02: Full-Stack React Web Development

course-react-web

Visuals: Built (30/30) | Duration: 21.4 min (18-25m Standard) | Allowed: component-tree, wireframe, flow, boxes, compare, states

Architectural Theme: Fiber Reconciliation, Hook Closures, Virtual DOM Diffing & Component Trees

Pedagogical Whiteboard Model: Show component trees with props cascading downwards and callback events bubbling up. Highlight component re-renders triggered by state setter dispatches.

Block	Days	Curriculum Topic & Real Lessons	Visual Engine Architecture
Block 1	Days 1–5	Declarative JSX parsing, Component composition & Virtual DOM tree	Template <code>component-tree</code> : Animated step-by-step state progression
Block 2	Days 6–10	useState hook closures, Immutable state updates & Re-render triggers	Template <code>wireframe</code> : Animated step-by-step state progression
Block 3	Days 11–15	Props contract interfaces, Prop drilling vs Context & Children nodes	Template <code>flow</code> : Animated step-by-step state progression
Block 4	Days 16–20	useEffect dependency arrays, Async fetch cycles & Cleanup timers	Template <code>boxes</code> : Animated step-by-step state progression
Block 5	Days 21–25	Controlled forms, Validation state machines & UI layout wireframes	Template <code>compare</code> : Animated step-by-step state progression
Block 6	Days 26–30	Next.js App Router hybrid rendering, Server vs Client components	Template <code>states</code> : Animated step-by-step state progression

Capstone Milestone: Production Enterprise SaaS Portal: High-performance Next.js application with dynamic dashboard routing, dark/light theme tokens, and live analytics telemetry.

Course 03: Node.js & TypeScript Backend Engineering

course-node-web

Visuals: Built (30/30) | Duration: 23.6 min (18-25m Standard) | Allowed: sequence, stack-queue, flow, boxes, table, states

Architectural Theme: Asynchronous Event Loop, Middleware Pipelines & Relational Microservices

Pedagogical Whiteboard Model: Map incoming HTTP request flows through Express middleware waterfalls. Show database connection pool acquisitions and async non-blocking I/O queues.

Block	Days	Curriculum Topic & Real Lessons	Visual Engine Architecture
Block 1	Days 1–5	V8 Engine Call Stack, libuv Event Loop & Microtask/Macrotask queues	Template <code>sequence</code> : Animated step-by-step state progression
Block 2	Days 6–10	TypeScript strict typing, Interface schemas & Compile-time validation	Template <code>stack-queue</code> : Animated step-by-step state progression
Block 3	Days 11–15	HTTP Request-Response lifecycles, RESTful routes & Express middleware	Template <code>flow</code> : Animated step-by-step state progression
Block 4	Days 16–20	Relational PostgreSQL queries, Connection pooling & ACID transactions	Template <code>boxes</code> : Animated step-by-step state progression
Block 5	Days 21–25	JWT HMAC-SHA256 authentication, Bcrypt password hashing & Auth guards	Template <code>table</code> : Animated step-by-step state progression
Block 6	Days 26–30	Containerized production Docker build & Health-check endpoint telemetry	Template <code>states</code> : Animated step-by-step state progression

Capstone Milestone: High-Concurrency REST Microservice: Authenticated backend API with role-based access control, connection-pooled PostgreSQL persistence, and Docker multi-stage deployment.

Course 04: Cloud Native Architectures (AWS)

course-cloud-native

Visuals: **Built (30/30)** | Duration: **24.1 min (18-25m Standard)** | Allowed: workflow, sequence, states, table, bars, compare

Architectural Theme: AWS VPC Topologies, Serverless Lifecycles & Highly Available Cloud Systems

Pedagogical Whiteboard Model: Visualize cloud architecture topologies: VPC subnets, Internet Gateways, Application Load Balancers, and ECS tasks. Clarify IAM role evaluation logic.

Block	Days	Curriculum Topic & Real Lessons	Visual Engine Architecture
Block 1	Days 1–5	AWS Global Cloud Infrastructure, Regions, Availability Zones & IAM Policies	Template workflow : Animated step-by-step state progression
Block 2	Days 6–10	VPC CIDR subnets, Public vs Private routing & Security Group firewalls	Template sequence : Animated step-by-step state progression
Block 3	Days 11–15	Compute scaling: EC2 instances, Application Load Balancers & Auto Scaling	Template states : Animated step-by-step state progression
Block 4	Days 16–20	Serverless computing: AWS Lambda triggers, API Gateway & DynamoDB NoSQL	Template table : Animated step-by-step state progression
Block 5	Days 21–25	Container orchestration: ECR image registries & ECS/Fargate task definitions	Template bars : Animated step-by-step state progression
Block 6	Days 26–30	Multi-AZ High Availability, Route53 DNS failover & Terraform IaC blueprints	Template compare : Animated step-by-step state progression

Capstone Milestone: Multi-AZ Serverless Cloud Infrastructure: Terraform-provisioned VPC architecture with ALB traffic distribution, ECS Fargate microservices, and automated health checks.

Course 05: DevOps & CI/CD Pipeline Automation

course-devops-cicd

Visuals: **Built (30/30)** | Duration: **22.6 min (18-25m Standard)** | Allowed: flow, workflow, states, sequence, compare

Architectural Theme: Immutable Containerization, Automated CI/CD Pipelines & Kubernetes Orchestration

Pedagogical Whiteboard Model: Diagram Docker layer caching trees, GitHub Actions stage execution waterfalls, and Kubernetes ReplicaSet pod reconciliation loops.

Block	Days	Curriculum Topic & Real Lessons	Visual Engine Architecture
Block 1	Days 1–5	POSIX Linux administration, Shell piping & Systemd daemon services	Template flow : Animated step-by-step state progression
Block 2	Days 6–10	Git trunk-based branching, Semantic versioning & GitOps workflows	Template workflow : Animated step-by-step state progression
Block 3	Days 11–15	Docker engine internals, Multi-stage image builds & Layer caching DAGs	Template states : Animated step-by-step state progression
Block 4	Days 16–20	GitHub Actions runners, Lint-Test-Build matrix workflows & Secret vaults	Template sequence : Animated step-by-step state progression
Block 5	Days 21–25	Kubernetes cluster architecture: Control plane, Pods, Services & Ingress	Template compare : Animated step-by-step state progression
Block 6	Days 26–30	Zero-downtime Blue/Green deployments, Canary rollouts & Prometheus alerts	Template flow : Animated step-by-step state progression

Capstone Milestone: Enterprise Automated GitOps Pipeline: Multi-stage Docker containerization pipeline with automated GitHub Actions testing, security vulnerability scanning, and Kubernetes deployment.

Course 06: UI/UX Design Systems & Visual Frontend

course-design-systems

Visuals: Needed (0/30) | Duration: 22.5 min (18-25m Standard) | Allowed: wireframe, component-tree, compare, boxes, bars

Architectural Theme: Atomic Component Architectures, Design Tokens & Accessible Web Layouts

Pedagogical Whiteboard Model: Draw CSS box-model margins/borders/padding, Flexbox/Grid spatial coordinates, and accessibility contrast verification ratios across light and dark modes.

Block	Days	Curriculum Topic & Real Lessons	Visual Engine Architecture
Block 1	Days 1–5	Visual hierarchy principles, Modular typography scales & 8pt grid geometry	Template wireframe : Animated step-by-step state progression
Block 2	Days 6–10	Color space tokens, HSL palette generation & WCAG AAA contrast ratios	Template component-tree : Animated step-by-step state progression
Block 3	Days 11–15	Atomic Design methodology: Atoms, Molecules, Organisms & Templates	Template compare : Animated step-by-step state progression
Block 4	Days 16–20	CSS Layout mechanics: Flexbox axes, CSS Grid fr tracks & Responsive breakpoints	Template boxes : Animated step-by-step state progression
Block 5	Days 21–25	Interactive component state machines: Hover, Focus-visible, Active & Disabled	Template bars : Animated step-by-step state progression
Block 6	Days 26–30	Design Token architecture: Style Dictionary, JSON token exports & Figma sync	Template wireframe : Animated step-by-step state progression

Capstone Milestone: Production Design System Library: Documented, accessible UI component kit featuring tokens, atomic components, and responsive mobile-to-desktop wireframe layouts.

Course 07: Data Structures & Algorithmic Optimizations

course-dsa-optim

Visuals: Built (30/30) | Duration: 23.3 min (18-25m Standard) | Allowed: cells, stack-queue, tree-graph, table, bars, compare

Architectural Theme: Asymptotic Complexity, Memory Buffers, Tree Traversal & Dynamic Programming

Pedagogical Whiteboard Model: Show memory array buffers with dual pointers shifting, recursion stack frame winding/unwinding, and 2D memoization grids populating optimal substructure solutions.

Block	Days	Curriculum Topic & Real Lessons	Visual Engine Architecture
Block 1	Days 1–5	Big-O Asymptotic Space-Time analysis, Linear arrays & In-place operations	Template cells : Animated step-by-step state progression
Block 2	Days 6–10	Two-pointer convergence, Sliding window bounds & Prefix sum caching	Template stack-queue : Animated step-by-step state progression
Block 3	Days 11–15	Linked memory lists, LIFO Call Stacks & FIFO Circular Queue buffers	Template tree-graph : Animated step-by-step state progression
Block 4	Days 16–20	Hierarchical Trees, Binary Search Tree balancing & Binary Heap priorities	Template table : Animated step-by-step state progression
Block 5	Days 21–25	Graph topologies, Adjacency lists, BFS/DFS traversal & Dijkstra shortest paths	Template bars : Animated step-by-step state progression
Block 6	Days 26–30	Dynamic Programming: Memoization tables, Tabulation & Knapsack optimization	Template compare : Animated step-by-step state progression

Capstone Milestone: High-Performance Algorithmic Optimization Suite: Benchmark-validated implementations of cache-conscious searching, graph pathfinding, and dynamic programming schedulers.

Course 08: Mobile Application Development

course-mobile-dev

Visuals: 🕒 **Needed (0/30)** | Duration: ⚠️ **6 min (Expansion to 18-25m Needed)** | Allowed: wireframe, component-tree, flow, sequence, states

Architectural Theme: Cross-Platform React Native, JSI Runtime, Native Modules & Offline SQLite

Pedagogical Whiteboard Model: Illustrate JavaScript-to-Native bridge execution, Yoga Flexbox layout reflows, and touch gesture responder state transitions.

Block	Days	Curriculum Topic & Real Lessons	Visual Engine Architecture
Block 1	Days 1–5	React Native architecture: JavaScript thread, JSI, Shadow Tree & Native UI	Template wireframe : Animated step-by-step state progression
Block 2	Days 6–10	Core UI primitives: View, Text, FlatList virtualization & Yoga flex layouts	Template component-tree : Animated step-by-step state progression
Block 3	Days 11–15	Navigation architectures: Native Stack, Bottom Tabs & Deep-link URI routing	Template flow : Animated step-by-step state progression
Block 4	Days 16–20	Offline state persistence: MMKV fast key-value & SQLite local transactions	Template sequence : Animated step-by-step state progression
Block 5	Days 21–25	Native hardware bridges: Camera capture, GPS Geolocation & Biometric auth	Template states : Animated step-by-step state progression
Block 6	Days 26–30	Performance profiling: Hermes bytecode compilation & Release bundle packaging	Template wireframe : Animated step-by-step state progression

Capstone Milestone: Cross-Platform Mobile Utility App: Feature-complete React Native application featuring virtualized infinite feeds, offline SQLite synchronization, and biometric authentication.

Course 09: Cybersecurity Principles & Secure Systems

course-cybersecurity

Visuals: ✅ **Built (30/30)** | Duration: 🕒 **24.4 min (18-25m Standard)** | Allowed: flow, sequence, table, compare, states

Architectural Theme: Cryptographic Protocols, OWASP Top 10 Mitigation & Zero-Trust Defense

Pedagogical Whiteboard Model: Diagram packet inspection sequences, TLS 1.3 cryptographic key exchanges, JWT tampering detection, and SQL injection sanitization barriers.

Block	Days	Curriculum Topic & Real Lessons	Visual Engine Architecture
Block 1	Days 1–5	Threat modeling methodologies, Attack vectors & OWASP Top 10 vulnerabilities	Template flow : Animated step-by-step state progression
Block 2	Days 6–10	Authentication architecture: Multi-factor auth, Salted Bcrypt & Session stores	Template sequence : Animated step-by-step state progression
Block 3	Days 11–15	Cryptographic systems: Symmetric AES-GCM, Asymmetric RSA & ECC key pairs	Template table : Animated step-by-step state progression
Block 4	Days 16–20	Web application hardening: CSP headers, CORS policies & CSRF token validation	Template compare : Animated step-by-step state progression
Block 5	Days 21–25	Network defense: TLS 1.3 handshake, WAF firewall rules & DDoS rate limiters	Template states : Animated step-by-step state progression
Block 6	Days 26–30	Security incident response: Penetration testing workflows & SIEM audit logs	Template flow : Animated step-by-step state progression

Capstone Milestone: Zero-Trust Secure Authentication Gateway: End-to-end hardened authentication microservice with Argon2id password hashing, rotating JWT signing keys, and brute-force rate limiting.

Course 10: Database Engineering & Query Performance

course-database-eng

Visuals:  Built (30/30) | Duration:  26.1 min (Expansion to 18-25m Needed) | Allowed: table, tree-graph, flow, compare, bars

Architectural Theme: Relational Theory, B+ Tree Indexes, Transaction Isolation & WAL Logging

Pedagogical Whiteboard Model: Illustrate B+ Tree node splits, Write-Ahead Log (WAL) sequential flushes, and query execution plans comparing sequential table scans with index lookups.

Block	Days	Curriculum Topic & Real Lessons	Visual Engine Architecture
Block 1	Days 1–5	Relational algebra, Entity-Relationship schemas & Normalization (1NF–BCNF)	Template table : Animated step-by-step state progression
Block 2	Days 6–10	SQL Query execution engine: Parsing, Query planning & AST optimization	Template tree-graph : Animated step-by-step state progression
Block 3	Days 11–15	Indexing data structures: B+ Tree page balancing, Hash indexes & Covering indexes	Template flow : Animated step-by-step state progression
Block 4	Days 16–20	ACID properties, MVCC concurrency control & Transaction isolation anomalies	Template compare : Animated step-by-step state progression
Block 5	Days 21–25	Query profiling: EXPLAIN ANALYZE interpretation & Buffer cache hit ratios	Template bars : Animated step-by-step state progression
Block 6	Days 26–30	Horizontal scaling: Table partitioning, Read replication & Connection pools	Template table : Animated step-by-step state progression

Capstone Milestone: High-Throughput Financial Ledger Database: Production-tuned PostgreSQL schema with B+ tree composite indexes, strict MVCC isolation, and sub-millisecond query execution plans.

Course 11: High-Scale Distributed System Design

course-distributed-sys

Visuals:  Built (30/30) | Duration:  22.8 min (18-25m Standard) | Allowed: sequence, workflow, states, table, cells, compare

Architectural Theme: Consensus Protocols, Consistent Hashing, Distributed State & Fault Tolerance

Pedagogical Whiteboard Model: Show consistent hashing token rings, Raft leader election heartbeat sequences, and 2-Phase Commit prepare/commit transaction message flows.

Block	Days	Curriculum Topic & Real Lessons	Visual Engine Architecture
Block 1	Days 1–5	Fallacies of distributed computing, Network partitions & CAP/PACELC theorems	Template sequence : Animated step-by-step state progression
Block 2	Days 6–10	RPC protocols, Protocol Buffers schema compilation & gRPC streaming channels	Template workflow : Animated step-by-step state progression
Block 3	Days 11–15	Data partitioning: Consistent hashing rings, Virtual nodes & Key distribution	Template states : Animated step-by-step state progression
Block 4	Days 16–20	Consensus algorithms: Paxos & Raft leader election, Term leases & Log commits	Template table : Animated step-by-step state progression
Block 5	Days 21–25	Distributed transactions: Two-Phase Commit (2PC) & Saga compensating patterns	Template cells : Animated step-by-step state progression
Block 6	Days 26–30	Resilience patterns: Circuit breakers, Distributed rate limiting & Jaeger tracing	Template compare : Animated step-by-step state progression

Capstone Milestone: Fault-Tolerant Distributed Key-Value Store: Raft-replicated cluster featuring automated leader failover, consistent hashing shard allocation, and vector clock conflict detection.

Course 12: AI Engineering & LLM Integration

course-ai-eng

Visuals:  **Built (30/30)** | Duration:  **22.6 min (18-25m Standard)** | Allowed: flow, table, bars, sequence, compare, boxes

Architectural Theme: Large Language Model Systems, Vector Similarity Search, RAG & Autonomous Agents

Pedagogical Whiteboard Model: Draw token embedding coordinate spaces, RAG vector retrieval cosine similarity rankings, and agentic tool dispatching request/response loops.

Block	Days	Curriculum Topic & Real Lessons	Visual Engine Architecture
Block 1	Days 1–5	Transformer architecture fundamentals: Tokenization, Embeddings & Self-Attention	Template flow : Animated step-by-step state progression
Block 2	Days 6–10	Prompt engineering patterns: Few-shot context, System framing & Chain-of-Thought	Template table : Animated step-by-step state progression
Block 3	Days 11–15	Vector databases: Dense embeddings, HNSW indexing & Cosine similarity search	Template bars : Animated step-by-step state progression
Block 4	Days 16–20	Retrieval-Augmented Generation (RAG): Document chunking & Context injection	Template sequence : Animated step-by-step state progression
Block 5	Days 21–25	Agentic tool calling: JSON function signatures, API dispatch & ReAct loops	Template compare : Animated step-by-step state progression
Block 6	Days 26–30	Evaluation & Guardrails: Hallucination detection, Token budgets & Model routing	Template boxes : Animated step-by-step state progression

Capstone Milestone: Enterprise Document Intelligence Copilot: Production RAG pipeline with semantic document chunking, hybrid vector search, citation synthesis, and guardrail verification.

Course 13: Full-Stack JavaScript Engineering

course-fullstack-js

Visuals:  **Needed (0/30)** | Duration:  **6 min (Expansion to 18-25m Needed)** | Allowed: flow, component-tree, sequence, table, states, compare

Architectural Theme: Monolithic to Microservice JS, Next.js Server Components, ORMs & WebSockets

Pedagogical Whiteboard Model: Show Next.js Server Component streaming hydration boundaries, WebSocket bidirectional frame ping-pong, and database mutation waterfalls.

Block	Days	Curriculum Topic & Real Lessons	Visual Engine Architecture
Block 1	Days 1–5	Modern ESNext language features, Microtasks, Async/Await & Generator functions	Template flow : Animated step-by-step state progression
Block 2	Days 6–10	Next.js 14 App Router: Server Components, Streaming Suspense & Route Handlers	Template component-tree : Animated step-by-step state progression
Block 3	Days 11–15	Server Actions, Client component boundaries & Optimistic UI updates	Template sequence : Animated step-by-step state progression
Block 4	Days 16–20	Type-safe ORM persistence: Prisma schemas, Relational joins & DB migrations	Template table : Animated step-by-step state progression
Block 5	Days 21–25	Real-time communication: WebSocket duplex channels & Server-Sent Events (SSE)	Template states : Animated step-by-step state progression
Block 6	Days 26–30	Production deployment: Edge middleware, Vercel serverless & Dockerized images	Template compare : Animated step-by-step state progression

Capstone Milestone: Real-Time Collaborative Workspace: Full-stack application with live multi-user WebSocket editing, Prisma database synchronization, and optimistic UI transitions.

Course 14: IoT, Firmware & Embedded Systems

course-iot-embedded

Visuals: ⌚ Needed (0/30) | Duration: ⚠ 6 min (Expansion to 18-25m Needed) | Allowed: flow, states, cells, boxes, sequence, register-bits

Architectural Theme: Hardware Registers, Memory-Mapped I/O, Interrupts & FreeRTOS Schedulers

Pedagogical Whiteboard Model: Diagram microcontroller memory-mapped registers, bitwise mask operations, interrupt vector dispatching, and FreeRTOS task priority preemption.

Block	Days	Curriculum Topic & Real Lessons	Visual Engine Architecture
Block 1	Days 1–5	MCU architectures (ARM Cortex-M), Memory map & GPIO register addressing	Template flow : Animated step-by-step state progression
Block 2	Days 6–10	Bitwise register manipulation: Bitmasking, Atomic Set/Reset (BSRR) & Pull-ups	Template states : Animated step-by-step state progression
Block 3	Days 11–15	Timers, PWM signal generation & Hardware Interrupt Service Routines (ISR)	Template cells : Animated step-by-step state progression
Block 4	Days 16–20	Analog-to-Digital Conversion (ADC): Quantization step sizing & Two-point scaling	Template boxes : Animated step-by-step state progression
Block 5	Days 21–25	Serial protocols: UART baud rates, I2C 7-bit bus addressing & SPI clock polarity	Template sequence : Animated step-by-step state progression
Block 6	Days 26–30	Real-Time Operating Systems: FreeRTOS preemptive scheduler, Queues & Semaphores	Template register-bits : Animated step-by-step state progression

Capstone Milestone: Embedded Industrial Telemetry Controller: FreeRTOS firmware implementing multi-channel ADC sensor acquisition, lock-free ring buffers, and I2C peripheral telemetry.

Course 15: 3D Interactive Graphics & Avatar Animation

course-3d-graphics

Visuals: ⌚ Needed (0/30) | Duration: ⚠ 6 min (Expansion to 18-25m Needed) | Allowed: tree-graph, boxes, flow, compare, bars

Architectural Theme: Vector Math, WebGL Render Pipeline, Shaders, Scene Graphs & Skeletal Rigging

Pedagogical Whiteboard Model: Visualize the 3D Model-View-Projection (MVP) matrix transformation pipeline, shader rasterization stages, and hierarchical skeletal bone transforms.

Block	Days	Curriculum Topic & Real Lessons	Visual Engine Architecture
Block 1	Days 1–5	3D Coordinate geometry: Cartesian vectors, Dot/Cross products & Matrix transforms	Template tree-graph : Animated step-by-step state progression
Block 2	Days 6–10	WebGL rendering pipeline: Vertex Shaders, Fragment Shaders & Rasterization	Template boxes : Animated step-by-step state progression
Block 3	Days 11–15	Three.js scene graph architecture: Object3D hierarchy, Meshes & Geometry buffers	Template flow : Animated step-by-step state progression
Block 4	Days 16–20	Material physics & Lighting: Ambient, Point, Directional & Physically Based Rendering	Template compare : Animated step-by-step state progression
Block 5	Days 21–25	Skeletal animation: Rigged bone hierarchies, Skinning weights & Keyframe tracks	Template bars : Animated step-by-step state progression
Block 6	Days 26–30	Rendering optimizations: Draw call batching, Frustum culling & LOD geometries	Template tree-graph : Animated step-by-step state progression

Capstone Milestone: Interactive 3D Avatar Animation Studio: WebGL-accelerated 3D viewport featuring custom PBR lighting shaders, skeletal bone transforms, and interactive orbital controls.

Course 16: Blockchain, Web3 & Smart Contracts

course-blockchain-web3

Visuals: 🕒 **Needed (0/30)** | Duration: ⚠️ **6 min (Expansion to 18-25m Needed)** | Allowed: tree-graph, stack-queue, flow, sequence, compare, states

Architectural Theme: Cryptographic Hashing, Merkle Trees, EVM Gas Mechanics & Smart Contracts

Pedagogical Whiteboard Model: Show Merkle tree cryptographic proof verification, EVM memory/stack gas consumption charts, and reentrancy attack state transition barriers.

Block	Days	Curriculum Topic & Real Lessons	Visual Engine Architecture
Block 1	Days 1–5	Cryptographic hash functions (SHA-256, Keccak), Asymmetric signatures & P2P networks	Template tree-graph : Animated step-by-step state progression
Block 2	Days 6–10	Blockchain ledger structure: Block headers, Merkle Patricia Tries & Consensus	Template stack-queue : Animated step-by-step state progression
Block 3	Days 11–15	Ethereum Virtual Machine (EVM): Execution stack, Storage slots & Gas accounting	Template flow : Animated step-by-step state progression
Block 4	Days 16–20	Solidity contract development: Value types, Access modifiers & Event emissions	Template sequence : Animated step-by-step state progression
Block 5	Days 21–25	Smart contract security: Checks-Effects-Interactions, Reentrancy & Integer safety	Template compare : Animated step-by-step state progression
Block 6	Days 26–30	Decentralized frontend: Ethers.js / Viem, Metamask wallet RPC & Token transactions	Template states : Animated step-by-step state progression

Capstone Milestone: Decentralized Escrow Protocol & dApp: Production Solidity smart contract with multi-signature release conditions, reentrancy guards, and reactive Web3 frontend.

Course 17: IoT Wireless Networks & Protocols

course-iot-network

Visuals: 🕒 **Needed (0/30)** | Duration: ⚠️ **6 min (Expansion to 18-25m Needed)** | Allowed: flow, sequence, states, table, compare

Architectural Theme: Wireless RF Topologies, BLE GATT Profiles, LoRaWAN Chirps & MQTT Brokers

Pedagogical Whiteboard Model: Map BLE GATT Service/Characteristic hierarchies, LoRaWAN chirp spread spectrum uplinks, and MQTT publish/subscribe broker message routing.

Block	Days	Curriculum Topic & Real Lessons	Visual Engine Architecture
Block 1	Days 1–5	Wireless RF propagation fundamentals, Frequencies (ISM bands) & Network topologies	Template flow : Animated step-by-step state progression
Block 2	Days 6–10	Bluetooth Low Energy (BLE): GAP advertising, GATT Services & Characteristics	Template sequence : Animated step-by-step state progression
Block 3	Days 11–15	IEEE 802.15.4 & Zigbee Mesh: Multi-hop routing & Self-healing network trees	Template states : Animated step-by-step state progression
Block 4	Days 16–20	LoRa & LoRaWAN: Chirp Spread Spectrum (CSS), Gateway uplinks & Join procedures	Template table : Animated step-by-step state progression
Block 5	Days 21–25	IoT Application protocols: MQTT QoS 0/1/2 handshakes, CoAP & Lightweight M2M	Template compare : Animated step-by-step state progression
Block 6	Days 26–30	Ultra-low-power optimization: Deep sleep modes, Duty cycling & Energy budgets	Template flow : Animated step-by-step state progression

Capstone Milestone: Low-Power LoRaWAN Environmental Gateway: Edge mesh network firmware routing environmental sensor telemetry through encrypted LoRaWAN packets to an MQTT cloud broker.

Course 18: Edge AI, DSP & TinyML Systems

course-iot-edge-ai

Visuals: ⌚ Needed (0/30) | Duration: ⚠ 6 min (Expansion to 18-25m Needed) | Allowed: flow, bars, cells, table, compare

Architectural Theme: Digital Signal Processing, Feature Extraction, INT8 Quantization & TinyML

Pedagogical Whiteboard Model: Show sliding window sensor buffers, Fast Fourier Transform (FFT) spectrograms, and INT8 quantized weight matrices mapped to MCU memory arenas.

Block	Days	Curriculum Topic & Real Lessons	Visual Engine Architecture
Block 1	Days 1–5	Edge intelligence paradigms, Microcontroller resource budgets (SRAM/Flash)	Template flow : Animated step-by-step state progression
Block 2	Days 6–10	Digital Signal Processing (DSP): Discrete filtering, Windowing & Fast Fourier Transform	Template bars : Animated step-by-step state progression
Block 3	Days 11–15	Time-series feature extraction: Spectral power, Zero-crossing rates & Peak detection	Template cells : Animated step-by-step state progression
Block 4	Days 16–20	Model optimization for edge: Post-training INT8 quantization & Weight pruning	Template table : Animated step-by-step state progression
Block 5	Days 21–25	TensorFlow Lite for Microcontrollers (TFLM): Memory arenas & Flatbuffer models	Template compare : Animated step-by-step state progression
Block 6	Days 26–30	Real-time inference pipeline: Accelerometer gesture recognition & Anomaly detection	Template flow : Animated step-by-step state progression

Capstone Milestone: Edge TinyML Gesture Recognition Engine: Microcontroller-deployed model classifying continuous IMU sensor streams in real-time under a 32 KB SRAM memory constraint.

Course 19: Industrial IoT Security & Device Lifecycle

course-iot-security

Visuals: ⌚ Needed (0/30) | Duration: ⚠ 6 min (Expansion to 18-25m Needed) | Allowed: flow, sequence, states, compare, table

Architectural Theme: Hardware Root of Trust, Secure Boot, Mutual TLS & Secure OTA Lifecycles

Pedagogical Whiteboard Model: Diagram secure boot cryptographic signature verification chains, mTLS certificate exchanges, and dual-bank OTA firmware flash memory rollback safety.

Block	Days	Curriculum Topic & Real Lessons	Visual Engine Architecture
Block 1	Days 1–5	Industrial IoT threat vectors, Purdue Enterprise Reference Model & Attack surfaces	Template flow : Animated step-by-step state progression
Block 2	Days 6–10	Hardware Root of Trust: Secure Elements (ATECC608), TPMs & Cryptographic coprocessors	Template sequence : Animated step-by-step state progression
Block 3	Days 11–15	Secure Boot architectures: Cryptographic hash validation & Anti-rollback eFuses	Template states : Animated step-by-step state progression
Block 4	Days 16–20	Identity & Transport security: X.509 Device certificates & Mutual TLS (mTLS)	Template compare : Animated step-by-step state progression
Block 5	Days 21–25	Secure Over-The-Air (OTA) updates: Dual-bank flash partitions & Rollback protection	Template table : Animated step-by-step state progression
Block 6	Days 26–30	Industrial communication security: Encrypted Modbus-TCP, OPC UA & IEC 62443	Template flow : Animated step-by-step state progression

Capstone Milestone: Hardened Industrial Edge Controller: Secure boot-verified firmware implementing mutual TLS cloud authentication and failsafe dual-bank OTA update recovery.

Course 20: Python Programming & Backend Systems

course-python-backend

Visuals: Built (30/30) | Duration: 22.6 min (18-25m Standard) | Allowed: flow, sequence, boxes, table, states, compare

Architectural Theme: Python Asynchronous Systems, FastAPI, Pydantic & Distributed Task Queues

Pedagogical Whiteboard Model: Illustrate *asyncio* event loop coroutine yields, *Pydantic* JSON schema validations, and *Celery* asynchronous task distribution across *Redis* workers.

Block	Days	Curriculum Topic & Real Lessons	Visual Engine Architecture
Block 1	Days 1–5	Advanced Python paradigms: Context managers, Decorators & Generator iterables	Template <code>flow</code> : Animated step-by-step state progression
Block 2	Days 6–10	Asynchronous programming: <i>asyncio</i> event loops, Coroutine tasks & Non-blocking I/O	Template <code>sequence</code> : Animated step-by-step state progression
Block 3	Days 11–15	FastAPI architecture: Path operations, Pydantic type models & Dependency Injection	Template <code>boxes</code> : Animated step-by-step state progression
Block 4	Days 16–20	Database persistence: SQLAlchemy 2.0 Async Session, PostgreSQL & Alembic migrations	Template <code>table</code> : Animated step-by-step state progression
Block 5	Days 21–25	Distributed task execution: Celery workers, Redis message brokers & Background tasks	Template <code>states</code> : Animated step-by-step state progression
Block 6	Days 26–30	Production engineering: Docker containerization, Pytest unit tests & Rate limiters	Template <code>compare</code> : Animated step-by-step state progression

Capstone Milestone: High-Performance Asynchronous FastAPI Service: Production microservice featuring async database connection pooling, background Celery processing, and OpenAPI contracts.

Course 21: Quantitative Engineering & Low-Latency Trading Systems

course-quant-systems

Visuals: Built (30/30) | Duration: 21.4 min (18-25m Standard) | Allowed: stack-queue, table, bars, flow, cells, compare

Architectural Theme: Limit Order Books, VWAP Slicing, Low-Latency Sockets & Microsecond Systems

Pedagogical Whiteboard Model: Show *Limit Order Book (LOB)* bid/ask price queues, execution algorithms slicing volume over time, and kernel-bypass zero-copy socket buffer flows.

Block	Days	Curriculum Topic & Real Lessons	Visual Engine Architecture
Block 1	Days 1–5	Market microstructure: Limit Order Books (LOB), Depth of market & Bid-Ask spreads	Template <code>stack-queue</code> : Animated step-by-step state progression
Block 2	Days 6–10	Financial timeseries analytics: Returns, Rolling volatility & Log price transforms	Template <code>table</code> : Animated step-by-step state progression
Block 3	Days 11–15	Algorithmic execution strategies: VWAP, TWAP, Implementation shortfall & Slippage	Template <code>bars</code> : Animated step-by-step state progression
Block 4	Days 16–20	Low-latency system engineering: Zero-copy buffers, Cache locality & Kernel bypass	Template <code>flow</code> : Animated step-by-step state progression
Block 5	Days 21–25	Risk modeling & Portfolio metrics: Value at Risk (VaR), Sharpe Ratio & Max Drawdown	Template <code>cells</code> : Animated step-by-step state progression
Block 6	Days 26–30	High-frequency backtesting engine: Event-driven matching simulation & FIX protocol	Template <code>compare</code> : Animated step-by-step state progression

Capstone Milestone: Microsecond Limit Order Book Matching Engine: High-frequency order matching engine simulating sub-millisecond price-time priority execution and real-time PnL tracking.

Course 22: Digital Accounting & Taxation (B.Com / BBA)

course-digital-accounting

Visuals: 🕒 **Needed (0/30)** | Duration: ⚠️ **6 min (Expansion to 18-25m Needed)** | Allowed: table, flow, compare, bars, ledger-sheet

Architectural Theme: Double-Entry Invariants, Ledger Balancing, GST Taxation & ERP Compliance

Pedagogical Whiteboard Model: Draw double-entry debit/credit ledger columns maintaining balancing equality, 3-way invoice matching flows, and GST tax credit cascading waterfalls.

Block	Days	Curriculum Topic & Real Lessons	Visual Engine Architecture
Block 1	Days 1–5	Accounting equation fundamentals: Assets = Liabilities + Equity & Business Entity rule	Template table : Animated step-by-step state progression
Block 2	Days 6–10	Golden rules of accounting: Real, Personal & Nominal accounts & Journal posting	Template flow : Animated step-by-step state progression
Block 3	Days 11–15	Ledger posting: T-Accounts balancing, Closing balances & Trial Balance checksums	Template compare : Animated step-by-step state progression
Block 4	Days 16–20	Financial reporting: Trading Account, Profit & Loss statement & Balance Sheet layout	Template bars : Animated step-by-step state progression
Block 5	Days 21–25	Taxation architecture: GST Input Tax Credit cascading & TDS statutory deductions	Template ledger-sheet : Animated step-by-step state progression
Block 6	Days 26–30	Modern ERP workflows: Automated bank reconciliations & Suspense account resolution	Template table : Animated step-by-step state progression

Capstone Milestone: Automated Corporate Accounting & GST Audit Engine: End-to-end digital accounting ledger with automated journal entries, balancing trial balances, and GST compliance validation.

Course 23: Business Finance & Investment Management (B.Com / BBA)

course-finance-investment

Visuals: 🕒 **Needed (0/30)** | Duration: ⚠️ **6 min (Expansion to 18-25m Needed)** | Allowed: table, bars, flow, compare

Architectural Theme: Time Value of Money, DCF Valuation, WACC Capital Structures & Portfolio Risk

Pedagogical Whiteboard Model: Visualize Discounted Cash Flow (DCF) timelines, DuPont 3-stage ROE breakdowns, and debt vs equity cost weights in WACC capital structure models.

Block	Days	Curriculum Topic & Real Lessons	Visual Engine Architecture
Block 1	Days 1–5	Time Value of Money (TVM): Present Value, Future Value & Compounding schedules	Template table : Animated step-by-step state progression
Block 2	Days 6–10	Capital budgeting evaluation: Net Present Value (NPV), IRR & Payback period hurdles	Template bars : Animated step-by-step state progression
Block 3	Days 11–15	Cost of Capital: Weighted Average Cost of Capital (WACC) & Optimal Debt-Equity mix	Template flow : Animated step-by-step state progression
Block 4	Days 16–20	Working capital dynamics: Cash Conversion Cycle, Operating cycle & Liquidity ratios	Template compare : Animated step-by-step state progression
Block 5	Days 21–25	Corporate Valuation: DCF terminal values, Multiples analysis & DuPont 3-stage ROE	Template table : Animated step-by-step state progression
Block 6	Days 26–30	Modern Portfolio Theory: Efficient frontier, Sharpe ratio & Beta risk sensitivity	Template bars : Animated step-by-step state progression

Capstone Milestone: Corporate Financial Valuation & Investment Model: Comprehensive DCF valuation model with dynamic WACC sensitivity matrices, scenario planning, and working capital forecasts.

Course 24: Business Analytics & Decision Intelligence (B.Com / BBA / MBA) course-business-analytics

Visuals: Needed (0/30) | Duration: 6 min (Expansion to 18-25m Needed) | Allowed: bars, table, flow, compare

Architectural Theme: Statistical Distributions, Pareto ABC Analysis, Regression & Executive KPIs

Pedagogical Whiteboard Model: Show descriptive distribution curves, Pareto 80/20 ABC classification breakdowns, and linear regression trendlines against executive scorecard KPIs.

Block	Days	Curriculum Topic & Real Lessons	Visual Engine Architecture
Block 1	Days 1–5	Data-driven decision framework: Descriptive statistics, Mean, Median & Standard deviation	Template bars : Animated step-by-step state progression
Block 2	Days 6–10	Data distributions: Normal curves, Skewness, Outlier detection & Interquartile range	Template table : Animated step-by-step state progression
Block 3	Days 11–15	Business KPI architecture: Balanced scorecards, Attribution & Performance variance	Template flow : Animated step-by-step state progression
Block 4	Days 16–20	Predictive analytics: Simple & Multiple OLS Linear regression & Trend forecasting	Template compare : Animated step-by-step state progression
Block 5	Days 21–25	Classification & Segmentation: Customer RFM analysis & Logistic regression odds	Template bars : Animated step-by-step state progression
Block 6	Days 26–30	Executive data storytelling: Automated KPI dashboards & Decision intelligence reports	Template table : Animated step-by-step state progression

Capstone Milestone: Executive Business Intelligence & Decision Dashboard: End-to-end analytics report synthesizing customer segmentation, revenue regression forecasts, and automated KPI alert metrics.

Course 25: Marketing & Brand Management (B.Com / BBA / MBA)

course-marketing-branding

Visuals: Needed (0/30) | Duration: 6 min (Expansion to 18-25m Needed) | Allowed: flow, compare, table, bars

Architectural Theme: Strategic Frameworks, Consumer Psychology, Brand Equity & PLC Lifecycles

Pedagogical Whiteboard Model: Map customer journey stages, Keller brand equity pyramid levels, BCG growth-share matrix quadrants, and omnichannel campaign ROI bar charts.

Block	Days	Curriculum Topic & Real Lessons	Visual Engine Architecture
Block 1	Days 1–5	Strategic marketing foundations: 4Ps Marketing Mix, STP & Porter's Five Forces	Template flow : Animated step-by-step state progression
Block 2	Days 6–10	Consumer psychology: Cognitive decision journey, Perception biases & Social proof	Template compare : Animated step-by-step state progression
Block 3	Days 11–15	Brand identity architecture: Brand personality, Archetypes & Keller's Brand Equity	Template table : Animated step-by-step state progression
Block 4	Days 16–20	Product portfolio management: Boston Consulting Group (BCG) Matrix & Product Life Cycle	Template bars : Animated step-by-step state progression
Block 5	Days 21–25	Integrated Marketing Communications (IMC): Omnichannel messaging & Content strategies	Template flow : Animated step-by-step state progression
Block 6	Days 26–30	Marketing performance measurement: Customer Lifetime Value (CLV) & Brand sentiment	Template compare : Animated step-by-step state progression

Capstone Milestone: Comprehensive Brand Strategy & Go-To-Market Blueprint: Multi-channel brand positioning framework complete with customer persona journeys, messaging matrices, and ROI forecasts.

Course 26: Digital Marketing & Growth Strategy (B.Com / BBA / MBA)

course-digital-marketing

Visuals: ⌚ **Needed (0/30)** | Duration: ⚠️ **6 min (Expansion to 18-25m Needed)** | Allowed: flow, bars, table, compare, funnel

Architectural Theme: Acquisition Funnels, Ad Auction Rankings, SEO Architectures & CAC/LTV Unit Economics

Pedagogical Whiteboard Model: Diagram conversion funnel drop-off waterfalls, ad auction rank bid vs quality score calculations, and CAC payback period bar comparisons.

Block	Days	Curriculum Topic & Real Lessons	Visual Engine Architecture
Block 1	Days 1–5	Digital acquisition landscape: Inbound vs Outbound & Customer acquisition funnels	Template flow : Animated step-by-step state progression
Block 2	Days 6–10	Search Engine Optimization (SEO): Crawl budgets, Technical SEO, On-page & Backlinks	Template bars : Animated step-by-step state progression
Block 3	Days 11–15	Search Engine Marketing (SEM): Google Ads auction mechanics, Quality Score & CPC bidding	Template table : Animated step-by-step state progression
Block 4	Days 16–20	Paid Social Advertising: Meta Ad pixel tracking, Lookalike audiences & ROAS targets	Template compare : Animated step-by-step state progression
Block 5	Days 21–25	Email marketing automation: Behavioral drip sequences, Open rate & Churn telemetry	Template funnel : Animated step-by-step state progression
Block 6	Days 26–30	Growth analytics: Multi-touch attribution models, A/B testing & CAC-to-LTV payback	Template flow : Animated step-by-step state progression

Capstone Milestone: Growth Marketing & Performance Campaign Architecture: Complete multi-channel digital acquisition campaign with SEO keyword architecture, ad auction bidding models, and CAC/LTV funnels.

Course 27: E-Commerce & Digital Business (B.Com / BBA / MBA)

course-ecommerce-digital-biz

Visuals: ⌚ **Needed (0/30)** | Duration: ⚠️ **6 min (Expansion to 18-25m Needed)** | Allowed: flow, table, sequence, bars, compare, funnel

Architectural Theme: SKU Variant Matrices, Volumetric Logistics, Checkout Gateways & Marketplace Scaling

Pedagogical Whiteboard Model: Illustrate e-commerce checkout conversion funnels, payment gateway authorization handshakes, and dimensional weight shipping cost calculations.

Block	Days	Curriculum Topic & Real Lessons	Visual Engine Architecture
Block 1	Days 1–5	E-Commerce business models: Direct-to-Consumer (D2C), B2B & Multi-vendor marketplaces	Template flow : Animated step-by-step state progression
Block 2	Days 6–10	Product catalog engineering: Parent-child SKU variants, Attributes & Inventory sync	Template table : Animated step-by-step state progression
Block 3	Days 11–15	Checkout optimization: Cart abandonment mitigation, Payment gateways & Webhooks	Template sequence : Animated step-by-step state progression
Block 4	Days 16–20	Supply chain fulfillment: 3PL logistics, Dimensional weight & Last-mile delivery	Template bars : Animated step-by-step state progression
Block 5	Days 21–25	Marketplace scaling: Amazon Buy Box algorithms, Seller metrics & Commission fees	Template compare : Animated step-by-step state progression
Block 6	Days 26–30	Cross-border digital commerce: Currency localization, Duties, Taxes & Compliance	Template funnel : Animated step-by-step state progression

Capstone Milestone: Scalable E-Commerce Operations Blueprint: Full operational framework for high-volume digital commerce, including SKU catalog architecture, checkout conversion funnels, and logistics routing.

Course 28: Entrepreneurship & Business Management (B.Com / BBA / MBA)

course-entrepreneurship-biz-mgmt

Visuals: ⌚ Needed (0/30) | Duration: ⚠️ 6 min (Expansion to 18-25m Needed) | Allowed: flow, table, compare, bars

Architectural Theme: Business Model Canvas, TAM/SAM/SOM Sizing, Cap Table Dilution & Runway Dynamics

Pedagogical Whiteboard Model: Show Business Model Canvas component linkages, bottom-up TAM/SAM/SOM concentric circles, equity dilution round-by-round cap tables, and monthly burn rate runway curves.

Block	Days	Curriculum Topic & Real Lessons	Visual Engine Architecture
Block 1	Days 1–5	Entrepreneurial mindset, Problem-Solution fit & Lean Startup iterative hypothesis	Template flow : Animated step-by-step state progression
Block 2	Days 6–10	Business Model Canvas (BMC): Revenue streams, Cost structures & Value propositions	Template table : Animated step-by-step state progression
Block 3	Days 11–15	Market opportunity sizing: Total Addressable Market (TAM, SAM, SOM) calculation	Template compare : Animated step-by-step state progression
Block 4	Days 16–20	Startup financial modeling: Unit economics, Burn rate & Cash runway forecasting	Template bars : Animated step-by-step state progression
Block 5	Days 21–25	Venture financing mechanics: Term sheets, Convertible notes, SAFEs & Cap table dilution	Template flow : Animated step-by-step state progression
Block 6	Days 26–30	Operational scaling: Hiring frameworks, Agile organizational design & Milestone gates	Template table : Animated step-by-step state progression

Capstone Milestone: Venture Creation & Investor-Ready Business Plan: Comprehensive startup plan featuring validated Business Model Canvas, bottom-up market sizing, unit economics, and 5-year pro forma financials.

Course 29: Sales, Customer Success & CRM (B.Com / BBA / MBA)

course-sales-crm-success

Visuals: ⌚ Needed (0/30) | Duration: ⚠️ 6 min (Expansion to 18-25m Needed) | Allowed: flow, table, states, bars, compare, funnel

Architectural Theme: BANT/MEDDIC Qualification, Sales Velocity, CRM Workflows & Customer Retention

Pedagogical Whiteboard Model: Diagram sales velocity formula components, MEDDIC qualification scorecards, lead-to-opportunity state progression, and customer health score distributions.

Block	Days	Curriculum Topic & Real Lessons	Visual Engine Architecture
Block 1	Days 1–5	Modern B2B sales methodologies: Consultative selling, Challenger sales & Inbound leads	Template flow : Animated step-by-step state progression
Block 2	Days 6–10	Opportunity qualification frameworks: BANT (Budget, Authority, Need, Timing) & MEDDIC	Template table : Animated step-by-step state progression
Block 3	Days 11–15	Sales pipeline mechanics: Deal stages, Probability weightings & Sales Velocity formula	Template states : Animated step-by-step state progression
Block 4	Days 16–20	CRM systems architecture: Lead routing workflows, Activity tracking & Pipeline automation	Template bars : Animated step-by-step state progression
Block 5	Days 21–25	Commercial negotiation: Objection handling frameworks, Value selling & Closing tactics	Template compare : Animated step-by-step state progression
Block 6	Days 26–30	Customer Success & Retention: Net Retention Rate (NRR), Churn telemetry & Health scores	Template funnel : Animated step-by-step state progression

Capstone Milestone: Enterprise B2B Sales & CRM Operational Playbook: Configured CRM pipeline architecture with automated MEDDIC qualification scoring, sales velocity tracking, and retention playbooks.

Course 30: Operations, Supply Chain & Business Compliance (B.Com / BBA / MBA)

course-operations-supplychain-compliance

Visuals: Needed (0/30) | Duration: 6 min (Expansion to 18-25m Needed) | Allowed: flow, table, bars, compare

Architectural Theme: SIPOC Value Stream Mapping, EOQ Inventory Models, Kraljic Matrices & Quality Control

Pedagogical Whiteboard Model: Draw SIPOC value stream process flows, Economic Order Quantity (EOQ) cost trade-off curves, Kraljic matrix supplier quadrants, and Six Sigma defect distributions.

Block	Days	Curriculum Topic & Real Lessons	Visual Engine Architecture
Block 1	Days 1–5	Operations strategy: Process architecture, Capacity planning & Bottleneck identification	Template flow : Animated step-by-step state progression
Block 2	Days 6–10	Value Stream Mapping: SIPOC diagrams, Lead time reduction & Waste elimination (Muda)	Template table : Animated step-by-step state progression
Block 3	Days 11–15	Inventory optimization: Economic Order Quantity (EOQ), Safety stock & Reorder points	Template bars : Animated step-by-step state progression
Block 4	Days 16–20	Strategic procurement: Kraljic purchasing portfolio matrix & Vendor SLA evaluations	Template compare : Animated step-by-step state progression
Block 5	Days 21–25	Total Quality Management (TQM): Six Sigma DMAIC cycle, Statistical process control & Kaizen	Template flow : Animated step-by-step state progression
Block 6	Days 26–30	Business compliance: Supply chain ethics, ISO certifications & Regulatory audit governance	Template table : Animated step-by-step state progression

Capstone Milestone: End-to-End Supply Chain Optimization Plan: Operations master plan featuring value stream mapping, automated EOQ inventory replenishment models, and supplier quality audits.

Course 31: AI & Digital Transformation for Business (B.Com / BBA / MBA)

course-ai-digital-transformation

Visuals: Needed (0/30) | Duration: 6 min (Expansion to 18-25m Needed) | Allowed: workflow, flow, table, compare, bars

Architectural Theme: Enterprise AI Value Equations, Robotic Process Automation & AI Governance

Pedagogical Whiteboard Model: Illustrate enterprise AI value creation frameworks, RPA bot automation process flows, AI governance risk-level classifications, and digital maturity phase roadmaps.

Block	Days	Curriculum Topic & Real Lessons	Visual Engine Architecture
Block 1	Days 1–5	Digital Transformation imperatives: Legacy modernization vs Digital-first paradigms	Template workflow : Animated step-by-step state progression
Block 2	Days 6–10	Enterprise AI Value Equation: Cost reduction, Revenue acceleration & Customer experience	Template flow : Animated step-by-step state progression
Block 3	Days 11–15	Robotic Process Automation (RPA): Workflow orchestration & Repetitive task bot automation	Template table : Animated step-by-step state progression
Block 4	Days 16–20	Generative AI workplace deployment: Departmental copilot adoption & Prompt standards	Template compare : Animated step-by-step state progression
Block 5	Days 21–25	AI Governance & Ethics: Bias auditing, Privacy protection, IP safety & EU AI Act compliance	Template bars : Animated step-by-step state progression
Block 6	Days 26–30	Organizational Change Management: Cultural readiness, Upskilling roadmaps & ROI dashboards	Template workflow : Animated step-by-step state progression

Capstone Milestone: Enterprise AI Transformation Roadmap & Governance Charter: Multi-year digital transformation strategic roadmap with departmental RPA automation blueprints and ethical AI policies.

Course 32: Computer Literacy, Digital Productivity & OS Fundamentals

course-computer-fundamentals

Visuals: **Needed (0/30)** | Duration: **6 min (Expansion to 18-25m Needed)** | Allowed: flow, boxes, tree-graph, table, compare

Architectural Theme: Von Neumann Architecture, POSIX File Permissions, CLI Pipelines & TCP/IP Networking

Pedagogical Whiteboard Model: Diagram Input-Process-Output CPU bus flows, Unix command pipe data streams, file system inode permission trees, and TCP/IP 4-layer packet traversals.

Block	Days	Curriculum Topic & Real Lessons	Visual Engine Architecture
Block 1	Days 1–5	Computer architecture: CPU registers, Arithmetic Logic Unit (ALU), RAM & Bus topology	Template flow : Animated step-by-step state progression
Block 2	Days 6–10	Operating Systems fundamentals: Kernel mode vs User mode, System calls & Process threads	Template boxes : Animated step-by-step state progression
Block 3	Days 11–15	POSIX Command Line: Terminal navigation, File permissions (chmod/chown) & Pipe redirection	Template tree-graph : Animated step-by-step state progression
Block 4	Days 16–20	Memory and storage hierarchy: L1/L2/L3 CPU Caches, Virtual memory paging & RAID arrays	Template table : Animated step-by-step state progression
Block 5	Days 21–25	Computer networking basics: TCP/IP 4-layer model, IPv4 subnetting, DNS & Port routing	Template compare : Animated step-by-step state progression
Block 6	Days 26–30	Digital security & System health: Process monitoring, SSH keys & Preventive maintenance	Template flow : Animated step-by-step state progression

Capstone Milestone: Systems Administration & Networking Verification Suite: Practical command-line and systems automation project demonstrating process monitoring, shell scripting, and network diagnostics.

Course 33: Everyday AI Literacy & Prompt Engineering

course-ai-prompt-literacy

Visuals: **Built (30/30)** | Duration: **21.2 min (18-25m Standard)** | Allowed: flow, compare, boxes, table, bars

Architectural Theme: Prompt Engineering Frameworks, Chain-of-Thought Reasoning & Fact Verification

Pedagogical Whiteboard Model: Show prompt before/after optimization comparisons, Chain-of-Thought reasoning step paths, temperature sampling spaces, and fact-checking verification scorecards.

Block	Days	Curriculum Topic & Real Lessons	Visual Engine Architecture
Block 1	Days 1–5	Generative AI mechanics: Probabilistic language modeling, Tokens & Prediction spaces	Template flow : Animated step-by-step state progression
Block 2	Days 6–10	Prompt structuring foundations: Context, Persona, Instruction & Constraint frameworks	Template compare : Animated step-by-step state progression
Block 3	Days 11–15	Advanced prompting techniques: Few-shot examples, Chain-of-Thought (CoT) & Decomposition	Template boxes : Animated step-by-step state progression
Block 4	Days 16–20	Multimodal AI capabilities: Text-to-image prompting, Audio transcription & Visual analysis	Template table : Animated step-by-step state progression
Block 5	Days 21–25	Workplace productivity automation: Research synthesis, Executive drafting & Data extraction	Template bars : Animated step-by-step state progression
Block 6	Days 26–30	Cognitive guardrails: Hallucination identification, Source grounding & Ethical evaluation	Template flow : Animated step-by-step state progression

Capstone Milestone: Master Prompt Engineering Playbook: Production-grade collection of verified, structured prompt templates for workplace research, communication, data analysis, and validation.

Course 34: Excel & Data Analysis Fundamentals

course-excel-data-viz

Visuals: ⌚ **Needed (0/30)** | Duration: ⚠️ **6 min (Expansion to 18-25m Needed)** | Allowed: table, cells, bars, compare, flow

Architectural Theme: 2D Coordinate Grids, Logical Evaluation, XLOOKUP Intersections & Executive Pivot Dashboards

Pedagogical Whiteboard Model: Illustrate 2D grid cell referencing, XLOOKUP array search traversals, Pivot Table multidimensional aggregations, and executive waterfall chart components.

Block	Days	Curriculum Topic & Real Lessons	Visual Engine Architecture
Block 1	Days 1–5	Spreadsheet grid architecture: 2D coordinate referencing, Absolute (\$) vs Relative locks	Template table : Animated step-by-step state progression
Block 2	Days 6–10	Statistical functions: SUM, AVERAGE, COUNT, MIN/MAX & Weighted average calculations	Template cells : Animated step-by-step state progression
Block 3	Days 11–15	Logical evaluations: Single IF, Nested IF statements & Multi-condition AND/OR formulas	Template bars : Animated step-by-step state progression
Block 4	Days 16–20	Text sanitization & Date calculations: TRIM, CLEAN, TEXTJOIN, CONCAT & DATEDIF math	Template compare : Animated step-by-step state progression
Block 5	Days 21–25	Lookup & Reference functions: Classic VLOOKUP vs Modern XLOOKUP & 2-Way INDEX-MATCH	Template flow : Animated step-by-step state progression
Block 6	Days 26–30	Data visualization & Reporting: Pivot Tables, Multi-dimensional slicing & Executive charts	Template table : Animated step-by-step state progression

Capstone Milestone: Executive Financial & Operational Excel Model: Fully automated corporate spreadsheet workbook featuring dynamic XLOOKUP models, Pivot Table aggregations, and executive visual dashboards.

Course 35: Git, GitHub & Version Control Basics

course-git-version-control

Visuals: ⌚ **Needed (0/30)** | Duration: ⚠️ **6 min (Expansion to 18-25m Needed)** | Allowed: tree-graph, flow, states, compare, boxes

Architectural Theme: Directed Acyclic Graphs, 3-Tier Git Areas, 3-Way Merges & Collaborative PRs

Pedagogical Whiteboard Model: Draw 3-tier Git architecture (Working directory -> Staging -> Repository), Directed Acyclic Graph (DAG) commit trees, and 3-way merge conflict marker resolutions.

Block	Days	Curriculum Topic & Real Lessons	Visual Engine Architecture
Block 1	Days 1–5	Version control architecture: Working Directory, Staging Area (Index) & Local Repository	Template tree-graph : Animated step-by-step state progression
Block 2	Days 6–10	Core versioning operations: Commit SHA hashes, Atomic commits, Git log & History inspection	Template flow : Animated step-by-step state progression
Block 3	Days 11–15	Branching dynamics: Pointer movements, Fast-forward merges & Feature branch isolation	Template states : Animated step-by-step state progression
Block 4	Days 16–20	Branch reconciliation: 3-way merge algorithms, Merge conflict resolution & Interactive rebase	Template compare : Animated step-by-step state progression
Block 5	Days 21–25	Remote collaboration: Git remotes, Push/Pull mechanics, Upstream tracking & Fetching	Template boxes : Animated step-by-step state progression
Block 6	Days 26–30	GitHub team workflows: Pull requests, Code review etiquettes, Branch protection & Tags	Template tree-graph : Animated step-by-step state progression

Capstone Milestone: Multi-Branch Collaborative Repository Simulation: Realistic engineering project demonstrating trunk-based branching, clean rebase histories, resolved merge conflicts, and peer-reviewed PRs.

Course 36: Professional Tech Communication & Interview Mastery

course-softskills-communication

Visuals: 🕒 **Needed (0/30)** | Duration: ⚠️ **6 min (Expansion to 18-25m Needed)** | Allowed: flow, compare, table, states

Architectural Theme: Minto Pyramid Structured Messaging, Active Listening, STAR Method & Tech Interviews

Pedagogical Whiteboard Model: Diagram Minto Pyramid top-down communication hierarchies, STAR method (Situation, Task, Action, Result) storytelling flows, and email tone before/after comparison tables.

Block	Days	Curriculum Topic & Real Lessons	Visual Engine Architecture
Block 1	Days 1–5	Foundations of technical communication: Clarity, Conciseness & The BLUF principle	Template flow : Animated step-by-step state progression
Block 2	Days 6–10	Structured thinking models: Minto Pyramid, MECE principle & Executive summaries	Template compare : Animated step-by-step state progression
Block 3	Days 11–15	Professional written correspondence: High-impact emails, Slack etiquette & Documentation	Template table : Animated step-by-step state progression
Block 4	Days 16–20	Verbal presentations: Agile standup updates, Technical demos & Slide design principles	Template states : Animated step-by-step state progression
Block 5	Days 21–25	Interpersonal dynamics: Active listening, Constructive feedback loops & Conflict de-escalation	Template flow : Animated step-by-step state progression
Block 6	Days 26–30	Interview mastery: The STAR method, Behavioral question responses & Salary negotiation	Template compare : Animated step-by-step state progression

Capstone Milestone: Executive Career Portfolio & Interview Masterclass: Comprehensive career communication asset package featuring structured technical project demos, STAR behavioral interview decks, and executive writing samples.

Course 37: Natural Language Processing & Computational Linguistics

course-nlp

Visuals: ✅ **Built (30/30)** | Duration: 🕒 **21.5 min (18-25m Standard)** | Allowed: letters, cells, table, flow, bars, tree-graph

Architectural Theme: Tokenization, TF-IDF Vectors, Word2Vec Semantics & Transformer Self-Attention

Pedagogical Whiteboard Model: Show token sequence sliding windows, word embedding vector spaces with cosine distance, and Transformer Query-Key-Value self-attention matrix products.

Block	Days	Curriculum Topic & Real Lessons	Visual Engine Architecture
Block 1	Days 1–5	Text preprocessing pipelines: Unicode normalization, Tokenization, Stemming & Lemmatization	Template letters : Animated step-by-step state progression
Block 2	Days 6–10	Statistical representation: Bag-of-Words, N-grams & Term Frequency-Inverse Document Frequency	Template cells : Animated step-by-step state progression
Block 3	Days 11–15	Distributed vector embeddings: Continuous Bag of Words (CBOW), Skip-gram & Word2Vec geometry	Template table : Animated step-by-step state progression
Block 4	Days 16–20	Sequence modeling: Recurrent Neural Networks (RNN), Hidden states & LSTM/GRU gating mechanisms	Template flow : Animated step-by-step state progression
Block 5	Days 21–25	Transformer architecture: Scaled dot-product Self-Attention, Multi-head matrices & Positional encodings	Template bars : Animated step-by-step state progression
Block 6	Days 26–30	Computational language applications: Named Entity Recognition (NER), Sentiment scoring & Summarization	Template tree-graph : Animated step-by-step state progression

Capstone Milestone: Production Natural Language Processing Engine: End-to-end computational linguistic pipeline performing text preprocessing, TF-IDF/Dense embedding extraction, and sentiment/NER classification.

6. Phased Implementation Roadmap for the 24 Standalone Courses

Phase	Scope & Objectives	Verification Gate & Deliverables
Phase 1: Curriculum Expansion	Expand 23 legacy pilot courses into 6-part LongLessons (1,200–1,500 words per day) to hit the 18–25 min standard.	<code>estimateLessonMinutes</code> outputs 18.0 to 24.5 min across all 30 days.
Phase 2: Visual Suite Generation	Author visual specifications for the 24 courses across 144 blocks (720 days) using the 5-Step Loop.	Gate v2 (Rules R1–R14) outputs 100% PASS with deterministic AST bindings.
Phase 3: Student Simulation	Run <code>simulate-student-view.mts</code> across all 1,110 days and 6,660 lesson parts.	0 broken bindings, 0 render crashes, 100% clean full-matrix simulation.

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